Reproducing the results from: Post Processing Recommender Systems with Knowledge Graphs for Recency, Popularity, and Diversity of Explanations

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The purpose of this report is to reproduce the results from the paper [Balloccu et al. 2022] and check their feasibility. In the following, we identify potential difficulties in reproducing the results and give recommendations on how to evade them. In addition, we challenge some of the results through a series of statistical tests before offering recommendations on how the paper could be improved with respect to reproducibility and statistical testing.

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1 Introduction

In the conference paper [Balloccu et al. 2022] a new recommender system to improve the quality of the explanation of recommendations without substantially decreasing the recommendation utility is proposed. The three properties they use to influence the explanation quality are: *interaction recency*, *shared entity popularity*, and *explanation type diversity*. The increase in explanation quality is achieved through re-ranking approaches. They conduct experiments on two data sets (movies: *MovieLens-1M*, music: *LastFM-1B*) to prove the viability of their approach. The results imply that using these properties for re-ranking does improve the explanation quality without decreasing recommendation utility.

The paper provides a series of figures and tables which we aim to reproduce given the code, data sets, and explanations supplied by the paper.

In the following section we explain our approach to reproduce and validate the results. We offer an overview of what is provided and what is missing to fully reproduce the exact results. Moreover, we document any difficulties we have encountered and how we evaded them to obtain

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the results discussed in the third section. In the third section we show our own results and compare them to those provided in the paper. We discuss eventual differences and offer suggestions on why results might differ. Finally, we conclude the report with the most important findings and a series of recommendations with respect to reproducibility.

2 Solution method

This section describes our approach for reproducing the results in the paper. Our approach includes inspecting the provided resources on three levels: we looked at the paper itself, we compared the pre-processing description of the data with the actually provided preprocessed data, and how easily results can be reproduced with the given code and/or explanations.

2.1 Paper

For the paper, we focused on the descriptions, data sources, and code. More specifically, we looked at the tables and graphs and compared their values to see if they match with the text argumentation. In addition, we studied how the experimental setup was described, trying to understand all the necessary components of the pipeline, as well as trying to find missing explanations which might lead to inconsistencies when reproducing results. Also, we scanned the paper for any links that lead us to code repositories or data sources for the original or the pre-processed data.

2.2 Pre-processing and data

For the data, we first tried to find any pre-processed data which we can use directly to recreate results. In any case, we intended to materialize the pre-processed datasets on our own to investigate potential discrepancies and verify that the pre-processing is described correctly in the paper (or any linked papers).

2.3 Code

At first we looked for any linked repositories in order to understand the pipeline by analyzing the code structure, which libraries are needed (and which versions), how the data is ingested, used, and ultimately processed etc. Also, we planed to look at any parameters necessary to run the program and examined if the settings are equal to the descriptions in the paper (or if the descriptions are missing). Finally, we intended to re-create the tables and graphs to examine any differences. If no code were provided, we would (as best as possible) try to implement it ourselves. For executing the experiments in this paper we relied on different software capable of running *Python* (e.g. *PyCharm*, *Jupyter*).

3 Results

In this section we provide an overview of the results from our solution method. Again, we look at the paper, the data, and the code in different subsections.

3.1 Paper

The paper provides links to a *Github* repository with the code used for creating the results in the paper. In addition, we have links to the original data and the pre-processed data. Both, the code and the data, are discussed in more detail in the subsequent subsections.

Analyzing the content of the paper, we found numerous tables and graphs presenting the results from processing the two data sets (MovieLens and LastFM). The tables are explained in detail in the text. However, we observed two problems here: First, we cannot (always) use the tables to verify the numbers in the text, as they are not rounded consistently. For instance, in *Table 3* from the paper numbers are rounded to two decimal points, whereas the numbers in the text have three

decimal points. Because of that, the percentage values calculated in the text cannot be verified either. Second, the percentages (and some other values provided) that can be verified do not match the data illustrated in the graphs or tables. For instance, in Figure 1 we can see *Table 4* from the paper. The table illustrates the explanation quality (EQ) received when optimizing the model for different properties (*R - recency*, *P - popularity*, *D - diversity*). The text in the paper states *"The gain in EQ, when we combined the three properties together, was substantial with respect to the original PGPR (+57.60%), the best performing single-property optimization (D-PGPR : 31.96%), and the best performing two-property optimization (DR-PGPR : +51.11%)"* [Balloccu et al. 2022]. Starting with the first number, when we take the base *EQ* value for $PGPR_{FM} = 1.07$ and assume an 57.60% increase we get $DPR\text{-}PGPR_{FM} = 1.07 * 1.576 = 1.686 \neq 1.76$. Similarly for the other two values we get $D\text{-}PGPR_{FM} = 1.07 * 1.3196 = 1.412 \neq 1.5$ and $DR\text{-}PGPR_{FM} = 1.07 * 1.5111 = 1.617 \neq 2.01$, respectively. This is just one of a few instances where we encountered this problem. We do not have an explanation for this.

Table 4: Explanation quality (EQ, LIR, EP, ETD).

	ML1M				LASTFM			
	EQ ↑	LIR ↑	SEP ↑	ETD ↑	EQ ↑	LIR ↑	SEP ↑	ETD ↑
PGPR	0.81	0.43	0.26	0.12	1.07	0.56	0.38	0.13
R-PGPR	1.37	**0.88**	0.37	0.12	1.50	**0.93**	0.43	0.14
P-PGPR	1.22	0.44	0.63	0.15	1.42	0.55	0.67	0.20
D-PGPR	1.05	0.43	0.33	**0.29**	1.50	0.55	0.47	**0.48**
DP-PGPR	1.37	0.44	**0.70**	0.22	1.56	0.55	0.68	0.32
PR-PGPR	**1.71**	0.86	**0.70**	0.14	1.80	0.86	0.50	0.43
DR-PGPR	1.59	0.84	0.46	**0.29**	**2.01**	0.86	**0.69**	0.46
DPR-PGPR	1.70	0.79	0.67	0.23	1.76	0.83	0.63	0.29

For each metric: best result in **bold**, second-best result underlined.

Fig. 1. Screenshot from *Table 4* provided in [Balloccu et al. 2022]

3.2 Pre-processing and data

The link to the *Github* repository in the paper provides a series of other links for downloading all the necessary, already pre-processed, data used to create the results. In addition, links to the original data are provided.

Unfortunately, however, the *LastFM-1B* dataset is no longer available in its original form due to licensing issues (source). For the ML1M dataset we tried downloading the original data and pre-processing it using the given dataset mapper in order to check whether *our* preprocessed data is the same (i.e. it yields the same results) as the originally preprocessed data. This dataset mapper did not work out of the box, some changes were required. However, after implementing these fixes and performing pre-processing, we got exactly the same results (more about them later).

Data splitting was done by timestamp, that is, data was sorted and most recent samples were taken as test set, the rest serving as training set. In that sense there is no need to check for any random state initialization (seed), as it would not change the outcome of the splitting.

3.3 Code

The code used to create the results is available in a *Github* repository. Python was used as programming language and a `requirements.txt` file is provided, describing the necessary libraries and their versions, but there is no specific restriction on which Python version to use (only > 3.6).

Except for a few minor changes (made to ease our experimentation), all provided code runs out of the box. However, this is not the only repository used, as in the repository itself more repositories

(that are not written by the authors themselves) are linked. These were used to run some baseline state-of-the-art algorithms that serve as benchmarks for performance comparisons.

When working with only the initial repository and the provided pre-processed data, it is possible to reproduce *Figure 2*, *Table 4* and *Table 5*. However, the reported values are not equal. This is surprising, as we worked with (supposedly) the exact same initial settings, versions and data provided in the repository. We cannot forget to mention the complete lack of descriptions regarding the chosen α values for the explanation quality weighting. In the paper, α is described as a tuning parameter that leverages the model's ability to focus on recommendation or explanation quality. Higher values of α favor better explanations at the cost of worse recommendations (at least in theory). The paper reported five different α values in *Figure 2*.

Similarly, descriptions on which *agent-top-k* was used are missing too. This parameter can be used to tune which paths are considered to be a part of an explanation (a path is a chain of relations in the underlying knowledge graph, for example: you listened to item A, which was composed by artist B, which also composed song C, thus I recommend song C to you). The given model's implementation supports two different *agent-top-k* values, which one was used we do not know.

AS a consequence we had to perform the experiments for all α values and all *agent-top-k* values. Needless to say, none of the results obtained did not match the paper. In Figure 3 we can see the results obtained when reproducing *Figure 2* from the paper when using the 25-50-1 *agent-top-k* configuration. Although the values themselves do not match, the paper's hypothesis stands up to scrutiny: models trained to provide explanations based on recency (R-PGPR) do exhibit improved linking interaction recency (LIR) metrics. The same applies to other metrics and models as well. Another interesting finding is that the recommendation quality metric (NDCG in this case) was improved in almost all cases, even when the model was favoring explanation over recommendation quality. This suggests that the actual baseline might have not been correctly or fully trained.

In Figure 2 we show the results obtained when reproducing *Table 4* and *Table 5*. As mentioned above, results do not match.

Explanation quality lastfm_25-50-1, α=0.3

	EQ	LIR	SEP	ETD
PGPR	1.005	0.085	0.79	0.13
R-PGPR	1.04	0.159	0.748	0.133
P-PGPR	1.203	0.086	0.93	0.186
D-PGPR	1.117	0.08	0.75	0.287
DR-PGPR	1.208	0.157	0.758	0.292
DP-PGPR	1.287	0.08	0.905	0.301
PR-PGPR	1.218	0.158	0.894	0.166
DPR-PGPR	1.268	0.156	0.877	0.234

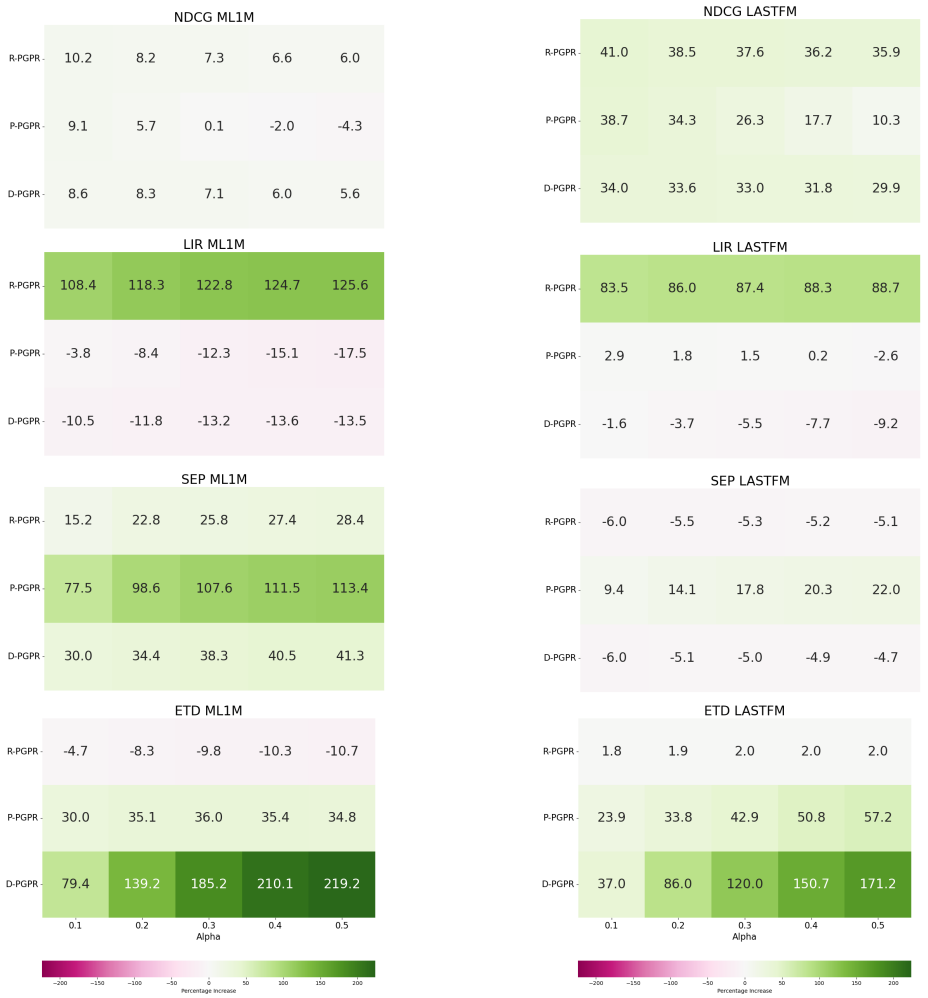
(a) Explanation Quality, LASTFM - 25-50-1

lastfm_25-50-1, α=0.3

	ΔNDCG	ΔLIR	ΔSEP	ΔETD
PGPR	-0.001	0.003	0.001	0.001
R-PGPR	0.001	0.013	0.002	0.001
P-PGPR	0.003	0.009	0.0	0.001
D-PGPR	-0.005	0.004	-0.0	0.001
DR-PGPR	-0.005	0.013	0.003	0.001
PR-PGPR	0.002	0.013	0.002	0.0
DP-PGPR	0.002	0.006	-0.001	0.002
DPR-PGPR	0.002	0.013	0.001	0.001

(b) Explanation Quality, LASTFM - 25-50-1

Fig. 2. Comparing explanation quality and fairness for LASTFM - 25-50-1



(a) ML1M - 25-50-1

(b) LASTFM - 25-50-1

Fig. 3. Reproduced Figure 2.

Working with the other repositories is not as easy, as they require adjustments to the code to make it run. Moreover, it requires to read into other cited papers to actually understand what is done. Running the code helps us to approximate the results in *Table 3*. The values we get are similar, but not equal. Presumably, this is due to differences in making the code run, as for the other repositories no exact descriptions are provided for procedures or parameters utilized.

Due to the provided data from the original repository providing questionable results, we also tried to retrain the PGPR-model 5 times per dataset with different seeds and run the optimization afterwards to see if anything changes. This proved to be a very time and resource intensive process, but led to interesting results. By recreating *Figure 2*, *Table 3*, *Table 4* and *Table 5* we compared our average results for all 5 runs per data set to the reported numbers. In contrast to the out-of-the-box results *Figure 2* now includes NDCG losses for increasing alphas as reported in the original paper, but with less drastic changes. The explanation quality changes also yield slightly different results

as reported. Results for *Table 3* are very similar to the reported numbers, but *Table 4* provides again different results with way higher explanation quality than reported for the *ML1M* dataset and way lower explanation quality for the *LastFM* dataset. *Table 5* proved to have less significant differences between male and female users with only the NDCG for the *ML1M* dataset really having problems with unfairness for our runs. More detailed results on the model re-runs can be found in *Stats.ipynb* in our *Github* repository.

4 Conclusion

In this final section we first provide an overview of our most significant learnings before offering a series of recommendations to improve the quality of the paper.

4.1 Learnings

Reproducing a scientific work should not be too difficult, both for the sake of science and practicality. But we have seen that reality most of the times is different than our ideals, and reproducing results from any given paper is not as trivial as it might have seem. We have learned an interesting lesson during this assignment, which is to be very cautious with our experimental setup and to keep in mind reproducibility from start to end of any scientific work.

4.2 Recommendations

We cannot guarantee that these recommendations will ensure reproducibility of any paper out there if followed, but they would have definitely helped in improving the one studied here:

- Cite your data sources (with all implications this leads to).
- Document every pre-processing done on the data.
- Freeze your environment: create a snapshot or provide explicit explanations on how to set up the environment in which you have conducted your experiments, as well as making sure your experimental setup works (pilot experiments).
- Explicitly state every decision that led you to reduce the number of possible outcomes.

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